

Certified Menus for Anonymous DHT Lookups

Distribution-free parameter selection with simultaneous guarantees under many-testing

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Abstract

Anonymous DHT lookups expose a familiar tension: stronger mechanisms (longer delegation walks, dummy queries, padding, retries) raise latency and bandwidth, so deployments tune parameters against traces. Unfortunately, tuning over many candidate configurations is a form of *many-testing*: after the fact, a surveillance adversary (or an unlucky deployment) can “pick the winner” among failed configurations, invalidating naive privacy claims.

We give a practical certify-and-select method with a distribution-free finite-sample guarantee that is *simultaneous* over an entire finite policy class. The output is a *certified menu*: a set of policies that provably satisfy anonymity, availability, and resource SLAs, together with a cost–privacy frontier that operators can switch between without re-running large experiments. We also show how to certify such menus off-policy from randomized shadow-routing logs and how to decouple expensive privacy labels from cheap telemetry via a two-sample protocol. This is an interface note and is intentionally compact; the companion disagreement-gated recertification paper (Certified C) develops the full deployment pipeline (anytime-valid monitoring, slice discovery, and change-control triggers) that consumes certified menus as inputs.

Scope. We isolate and formalize *Certified Menus for Anonymous DHT Lookups* as a reusable contract surface in the anonymity / Anonymous-DHT series.

Imports. This paper owns the many-testing-safe *selection* wrapper. We treat (i) what privacy labels mean as imported (Anonymity A; Synthesis 5),[?, ?] (ii) state-dependent pitfalls as imported (AnonDHT State 1),[?] and (iii) notarization mechanics as imported (Evaluation series). Menus are exported as contract objects (Synthesis 0) and should be referenced by ABOM/OINL manifests (Anonymity C).[?, ?]

Exports. A *certified menu* object (family of transcript projections + simultaneous privacy bounds) and a verifier-facing interface for publishing menu-level guarantees under repeated use.

Artifact policy. This archive is source-only; supporting artifacts are available on request. See [?].

1 Problem statement

Let $\Pi = \{\pi_1, \dots, \pi_M\}$ be a finite set of lookup policies. Each policy induces a distribution over transcripts and therefore over a chosen privacy label $L(\pi)$ (e.g., an odds-inflation bound for a projection) and over performance metrics such as latency $T(\pi)$ and bandwidth $B(\pi)$. We want to publish a *switchable* set of policies that satisfy:

$$\text{Privacy}(\pi) \leq \tau_P, \quad \text{Latency}(\pi) \leq \tau_T, \quad \text{Bandwidth}(\pi) \leq \tau_B,$$

simultaneously for all policies we certify, with a single overall failure probability α .

Because the policy pool can be large, post-hoc tuning is a form of *multiple testing*: a failed configuration can be “found” after the fact unless the certificate controls error uniformly over the whole pool. We therefore design the certificate as a family-wise (or simultaneous) guarantee using standard multiplicity control (e.g., Bonferroni/Holm-style corrections) rather than a single-policy bound.[?]

2 Certified menus

Definition 1 (Certified menu). *A certified menu at level α is a subset $\hat{\Pi} \subseteq \Pi$ produced from data such that with probability at least $1 - \alpha$, every $\pi \in \hat{\Pi}$ satisfies the published SLAs.*

The key is that the guarantee is *uniform over the output set*. This blocks the “pick the winner” failure mode.

2.1 A simple, distribution-free construction

Assume for each policy π_m we can compute a per-policy valid upper confidence bound (UCB) for each SLA quantity from n i.i.d. trials:

$$\Pr [\text{Privacy}(\pi_m) \leq \text{UCB}_P(m)] \geq 1 - \alpha_P(m)$$

(and similarly for latency/bandwidth). These can be simple: e.g., for bounded witnesses use Hoeffding; for Bernoulli “failure” events use Clopper–Pearson.

Theorem 1 (Union-bound menu validity). *If $\sum_{m=1}^M (\alpha_P(m) + \alpha_T(m) + \alpha_B(m)) \leq \alpha$ and we define*

$$\hat{\Pi} := \{\pi_m : \text{UCB}_P(m) \leq \tau_P, \text{UCB}_T(m) \leq \tau_T, \text{UCB}_B(m) \leq \tau_B\},$$

then $\hat{\Pi}$ is a certified menu at level α .

Remark 1 (Effective multiplicity). *This theorem is intentionally crude; Certified B shows how to replace M by an effective multiplicity M_{eff} when many policies share the same observable behavior.*

2.2 Switching without re-certifying

A practical menu includes a cost frontier: operators might switch among policies in $\hat{\Pi}$ as load varies. Because the guarantee is uniform over the entire certified set, switching does not invalidate the claim.

3 Off-policy certification from shadow routing

Often we have logs from a *randomized* policy π_0 that “explores” a superset of routes/actions. If for each candidate policy π_m we can compute an importance weight $w_m(X)$ for a transcript sample X (with bounded variance), then standard off-policy estimators produce valid bounds. We treat the details as a deployment choice; the only requirement for the menu theorem is that each UCB be valid at its allocated level.

4 Two-sample decoupling: expensive labels, cheap telemetry

Many privacy labels require heavyweight analysis (e.g., reconstructing committee contacts or stop times). A pragmatic trick is to certify a cheaper proxy statistic Y on every run and only compute X (the full label) on a subsample. A two-sample protocol checks that Y predicts X with sufficient reliability to support certification. (Operationally: publish both a label and a proxy audit bound in the receipt.)

5 Output format

We recommend publishing: (i) the certified menu policies (hashes/IDs), (ii) their UCBs and thresholds, and (iii) the multiplicity accounting (how α was split). ABOM/OINL (Anonymity C) is a good carrier for these fields.

6 Takeaways

- **Selection is part of the threat model:** tuning across many configurations is a many-testing problem, so publish *simultaneous* guarantees over a finite policy class rather than a single post-hoc “best” policy.
- **Menus are switchable:** once a set is certified uniformly, operators can switch among policies in that set without re-running expensive experiments.
- **Receipts should carry the multiplicity accounting:** the guarantee lives in how α was split (or compressed by effective multiplicity), so that split belongs in the published contract object (ABOM/OINL or an evaluation receipt).

References

- [1] Anonymous. Artifact and Disclosure Policy for the One-True-Archive (Source-Only Public Release). Series synthesis note (Synthesis 6), 2026. (This archive.)
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